# In-Distribution Gains Do Not Transfer: Leakage-Free Evaluation of Retrieval Interventions for Low-Resource Legal Question Answering

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***Abstract*—Retrieval interventions for legal question answering are almost always reported as a single number on a single evaluation set, and that number is almost always measured on questions drawn from the same distribution the system was tuned on. We report what happens when it is not. Fine-tuning a multilingual retriever on $4{,}692$ verified citation pairs for Pakistani law gains $+26.5$ Hit@1 on held-out questions from its own generator and $+2.4$ when the question style changes. A second model, trained on a tenth as much data from a different source, initially shows a $+16.5$ in-distribution gain, but target leakage invalidates that comparison; on clean out-of-distribution evaluation it changes by $-0.3$. Sorting every retrieval intervention we measured by whether its effect survived a change of question style produces a clean split: those encoding legal or document structure transferred, those learning question phrasing did not, including a general-purpose cross-encoder that moved the correct statute from rank 3 to rank 10 where a deterministic statutory-scope rule placed it first in under a millisecond. We further report three evaluation defects—an index containing its own test answers, a systematically wrong gold-label join, and target leakage between train and test—each of which inflated or distorted a headline number until it was checked, and each time the flattering version was the wrong one; none was visible in the metrics, and in each case the correction removed a result we had already believed. Throughout, the measurement protocol determined the conclusions more than any technique did.**

***Index Terms*—legal informatics, retrieval-augmented generation, retrieval evaluation, distribution shift, data contamination, low-resource NLP, code-switching, Urdu**

## I. Introduction

Access to legal information is unevenly distributed. In Pakistan, where legal counsel is scarce outside major urban centres and court backlogs leave disputes unresolved for years, a litigant who cannot afford counsel is often unable to establish even the threshold facts of their own position: which statute governs, which forum has jurisdiction, what a filing will cost, or whether a limitation period has already expired.

Retrieval-augmented generation [1] is the standard response, and legal question answering is now an active application area [2], [3]. Applying it to Pakistan is not a matter of swapping the corpus. Queries arrive in English, in Urdu script, and most commonly in Roman Urdu, the transliterated register used in everyday digital communication; the applicable law combines codified statute with Islamic personal law and varies by province; and users routinely import Indian statutory terminology, asking about the Indian Penal Code when the Pakistan Penal Code governs. Each is a retrieval problem before a generation problem, and none has an established benchmark.

**Reproducibility.** All evaluation code, corpus ingestion manifests, contamination detection scripts, and fine-tuning configurations are publicly available to enable independent auditability.

That absence is the subject of this paper. When no independent evaluation set exists, a system's developers build one, tune against it, and report the resulting number, and the literature offers little guidance on how much such a number is worth. We set out to improve retrieval for this jurisdiction and measured each intervention twice: once on questions from the distribution it was developed against, and once on questions written by a different generator for a different purpose. The gap between those two measurements turned out to be larger than the gap between any two techniques we tried.

We report it as a negative result with a positive structure. Interventions divide by *what they encode*: those encoding a property of the law or the documents survived the change of question style, while those learning a mapping from question phrasing to passage did not, however large their in-distribution gain. The same division predicts which of our corpus repairs held.

**Scope of claims.** Several results below are defects we found in our own system, reported as findings rather than apology: each was invisible to the metric meant to catch it, and these failure modes are not specific to a single model configuration. What we do *not* claim is equally definite: no end-to-end improvement in answer correctness, and no Pakistan-wide coverage. Section VII states each limit and what it bears on.

The system that produced these measurements is described only as far as the results require (Section III); its architecture is reported separately. Our contributions are:

- **A transfer measurement of retrieval fine-tuning**, over two models trained independently at a tenfold difference in scale, with an account of why that predicts which interventions transfer (Sections V-B, VI-A).
- **Three contaminations and how each was found**: a leaking index, a systematically wrong gold-label join, and target leakage between test and training, none visible in the metrics (Section V-D).

- **Verified citation supervision** for a jurisdiction with no query/passage training data, built from citations because a citation is checkable (Section IV-A).

## II. Related Work

**Legal NLP and retrieval-augmented generation.** Domain-adapted encoders such as LEGAL-BERT [2] established that legal text benefits from in-domain pretraining, and LegalBench [3] benchmarks legal reasoning in large language models. Retrieval augmentation [1] over dense passage retrieval [4] is the standard architecture for grounding answers in sources, with Self-RAG [5] adding self-reflection over retrieval quality. This line of work targets English common-law or EU material almost exclusively, and reports in-domain evaluation on benchmarks it did not construct, a luxury this jurisdiction does not afford.

**Low-resource and hybrid retrieval.** Multilingual dense encoders [6] make a single index viable across scripts, and hybrid lexical–dense retrieval with rank fusion [7], [8] remains strong where exact terminology matters, as it does for statute references. Neither line reports what fusion does to a dense channel fine-tuned on the domain, which Section V-C measures.

**Generalisation and contamination in evaluation.** That a retriever's gain may not survive a change of domain is established for English: BEIR [9] evaluates retrievers zero-shot across eighteen datasets and MTEB [10] extends the idea to embedding models generally, both finding in-domain strength a poor predictor of out-of-domain strength. Neither has an analogue for a low-resource jurisdiction, where the practical difficulty is that a developer must build the only evaluation set that exists. Contamination is likewise recognised as a systematic threat to reported results [11], [12], though that literature concentrates on pretraining corpora absorbing benchmark text; we report the retrieval-specific case, in which the *index* is itself a system component and can leak (Section V-D).

What none of these lines reports, and what we report here, is whether a retrieval intervention's measured gain survives a change in who wrote the questions.

## III. Experimental Setup

### A. Corpus and Retrieval

The statute corpus is drawn from a public collection of 969 Pakistani legal documents [13], itself converted from Ministry of Law and Justice publications and distributed under ODC-BY; we segment it by section and partition it into four collections by legal domain (civil, criminal, family, constitutional). It is supplemented with instruments fetched directly from the official consolidated legislation service [14], including the Code of Civil Procedure 1908 and twenty family-law statutes that the public collection omitted. The constitutional collection holds article text authenticated against the National Assembly publication; a bilingual English–Urdu constitutional question–answer set [15] supplies evaluation questions only and is *not* indexed, for reasons Section V-D reports. A separate corpus of reported judgments from six courts supports case-law retrieval. Figure 1 reports the indexed chunk counts. The sources are government publications under an attribution licence, given here and in the released ingestion code.

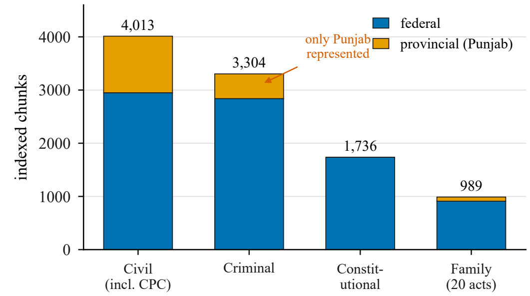


Fig. 1. Indexed statutory corpus by collection and legislative tier, with a separately indexed judgment corpus of 7,529 chunks from six courts not shown. Devolved provincial coverage is Punjab-only; constitutional law is inherently federal.

Retrieval is hybrid. A BM25 index and a dense index over `multilingual-e5-base` embeddings are combined by weighted ensemble with weights 0.6 and 0.4 respectively, the lexical channel weighted higher because exact statute and section references carry disproportionate signal in this domain. Dense retrieval applies the asymmetric `query:`/ `passage:` prefixes the model expects. Both channels are filtered so that a query for a given province retrieves only that province's provisions or federal ones: the dense channel by metadata predicate, the lexical channel by post-retrieval filter, since BM25 admits no index-time constraint.

That filter is a mechanism, not a coverage claim: of 10,042 statutory chunks, 1,608 are provincial and all of them are Punjab (Figure 1), so a query from any other province is served from federal law alone. Since tenancy, land revenue and local taxation are devolved, the jurisdictions absent are those that would need the filter most. A rule-based alias layer additionally rewrites cross-jurisdictional references, users frequently naming Indian codes when Pakistani ones govern.

## IV. Evaluation Protocol

Rather than authoring question–answer pairs, we label recorded traffic: judgements are pooled to a fixed depth, stored with it, and metrics below the pool refused as unjudged rather than counted irrelevant. Each turn carries one of four verdicts (correct, incorrect, correct refusal, wrong refusal), since collapsing a correct abstention into a mistaken one loses the distinction the system exists to make. That set is not yet of sufficient volume to report (Section VII); what follows is used for everything that is.

### A. Verified Citation Supervision

Fine-tuning a retriever needs query/passage supervision, which for most of this jurisdiction does not exist. We derive it from a public corpus of 11,195 Pakistani legal question/answer pairs carrying machine-parseable citations (`PPC s.467`, `CONST art.203F`) and domain labels. The answers are LLM-generated; the *citation* is what we use, because a citation is a checkable label and an answer is not. Nothing from this source is indexed.

Four filters are applied in order, and their yield is reported because the rejection rate is itself informative. Rows seeded from LEGAL-UQA are dropped (911), since that dataset supplies our evaluation set. Citations naming a section the corpus does not hold are dropped (720), including all 208 civil-procedure rows before the Code of Civil Procedure was ingested. Each survivor is then *validated*: the cited section must share rarity-weighted vocabulary with the answer, since a confidently wrong citation trains the retriever to reproduce the error (203 dropped), and duplicates removed. 5,883 pairs survive (52.6%), split 80/20 into 4,692 training and 1,191 evaluation pairs; 32% pair Urdu-script queries with English statute, the code-switched case for which no training data previously existed.

Validation is deliberately lexical rather than embedding-based: scoring candidates with the base model and keeping those it already agrees with would discard the hard examples that carry signal, and would flatter the fine-tuning that follows. It also proved an effective corpus check. It flagged citations to Article 9 as topically inconsistent, and the citations were right: the corpus was wrong, indexing 164 Schedule paragraphs as Articles 1–8, which include the fundamental rights and the high-treason provision. The index had been serving Schedule text under the citation of a constitutional right.

### B. Metrics

Retrieval is reported by Hit@$k$, mean reciprocal rank and nDCG@$k$ [16], at cut-offs within the pooling depth. We note one implementation detail because it cost us a result: nDCG's ideal-DCG denominator must be computed from $|G_q|$, and a fixed denominator of 1, correct only for single-gold queries, silently deflates the multi-gold queries that matter most. It was a defect in one of our own evaluation scripts.

## V. Results

Three of the four collections changed materially during this work, for reasons the evaluation surfaced rather than planned: the constitutional collection was rebuilt after 67% of it proved to be generated text (Section V-D), the Code of Civil Procedure 1908 was found absent entirely, and family law, the smallest collection, and the one where a wrong answer does most harm, grew from 193 to 989 chunks once twenty statutes listed in the project's own fetch manifest were actually downloaded. Corpus defects, not model capacity, accounted for the largest single improvements we made.

Section VII states plainly what remains unmeasured.

### A. Cross-Encoder Reranking: A Negative Result

Cross-encoder reranking [17] is the standard remedy for the symptom we faced, the correct statute retrieved but ranked below a lexically similar competitor, so we measured it rather than assuming it. The decisive case is the following: whether a tenant may be evicted without notice in Punjab, where the correct instrument is the Punjab Rented Premises Act 2009 (urban rental) and the competitor the Punjab Tenancy Act 1887 (agricultural tenancy). Both are genuinely about tenants and eviction, so this is an ambiguity of statutory scope rather than a vocabulary gap, and it appeared only as the corpus grew to include provincial law.

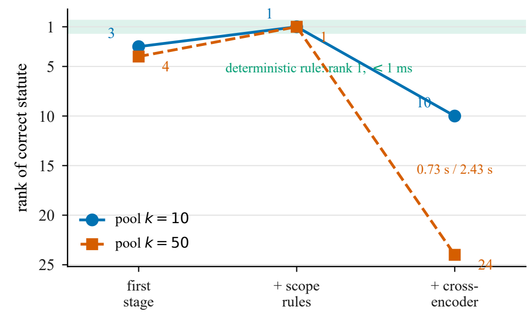


Fig. 2. Rank of the correct statute (Punjab Rented Premises Act 2009) after each stage, at both retrieval depths; lower is better. The deterministic statutory-scope rule places it first at both depths in under a millisecond, where the cross-encoder demotes it and grows worse as the pool widens.

Figure 2 gives the result. The cross-encoder does not merely fail to help; it moves the correct statute *away* from the top, promoting the agricultural statute at both depths, the confusion it was introduced to resolve. Degradation growing with pool size is the signature of a scoring function ordered systematically against the target rather than merely noisy.

Reporting this as a uniform failure would overstate it. Across six queries the reranker improved three (the *khula* target from rank 8 to 4, an FIR-registration query from 2 to 1, a theft query from 3 to 1), left two unchanged and badly worsened one, but the one it worsens is the case that motivated it, and those it improves were already nearly correct. It sharpens rankings that did not need sharpening while inverting the one that did.

We attribute this to train/test distribution shift rather than model capacity. The checkpoint is trained on MS MARCO, whose relevance judgements concern short web passages answering informational queries, whereas the distinction here, whether *tenant* means a cultivator under an 1887 revenue statute or an occupant of urban premises under a 2009 one, is one of legislative scope. A relevance model trained on topical overlap has no representation of which instrument *governs*. Prior work reports the same direction of effect, attributing degraded legal case retrieval to domain mismatch on long legal text unlike the reranker's training data [18]; that this reproduces on statutory retrieval in a different jurisdiction and language setting indicates that these failure modes are not specific to a single model configuration.

Widening first-stage retrieval from 10 to 50 candidates was reverted alongside it: without an effective reranker, additional depth is additional noise. A legal-domain cross-encoder may well succeed where the generic model failed; the finding is that the off-the-shelf model is actively harmful on this text, not that reranking is unsound.

### B. Domain Fine-Tuning: Large In-Distribution Gains That Do Not Transfer

The diagnosis of Section V-A left a specific gap. Across the total corpus asset of 592 bilingual legal questions (LEGAL-UQA), initial dense ablation was evaluated on an in-distribution subset of 200 held-out constitutional questions (where dense Hit@50 reaches 89% but Hit@1 reaches only 43.5%, a 45-

TABLE I
THREATS TO VALIDITY: WHAT EACH BEARS ON AND ITS STATUS.

| Threat | Bears on | Status |
|---|---|---|
| Evaluation questions are developer-authored or LLM-generated | external validity of every Hit@$k$ figure reported | 3,852 independently sourced questions identified; two-annotator labelling with a frozen holdout is the first item of Section VI-C |
| Provincial coverage is Punjab only (1,608 of 10,042 chunks) | any Pakistan-wide claim; devolved subjects most of all | claims are restricted to federal law plus Punjab throughout (Section III-A) |
| Generation correctness is unmeasured | end-to-end benefit to a user | all claims are retrieval-level; practitioner adjudication is the measurement we would prioritise (Section VI-C). One recorded instance indicates the cost is not hypothetical: for the query *"What is the punishment for theft under the Pakistan Penal Code?"* the system cited PPC 382 (theft after preparation for causing hurt) rather than PPC 379 (theft), and attributed procedure to a non-existent provincial Criminal Procedure Code. The turn was recorded with `bm25_confidence` $= 0.0$ and retrieval relevance $0.2396$, yet was reported at confidence $0.85$, marked grounded, and returned. The failure is upstream of generation: query normalisation rewrote the English question into corrupted Urdu, injecting the token *sarkari* ("official") three times despite a specification requiring English queries to pass through unchanged, so retrieval searched a different concept and returned no chunk of PPC 379. Signals that would have supported abstention were present and unheeded |
| The scope-rule and reranking result rests on one worked case | how far the rank $3 \rightarrow 10$ demotion generalises | reported as a mechanism with a named cause, corroborated independently [18], not as a rate over queries |
| The initial provenance pool was authored alongside the system | any judgement drawn from those turns, and the abstention splits they define | the seed queries are hand-written, not system-generated, and are marked as such; traffic recorded after the seeding stage is independent, and the distinction is carried in the record rather than reconstructed later |

point ranking deficit). For cross-style generalisation, evaluating the general model (trained on 4,692 citation pairs) required filtering the 592-question set for target leakage, leaving 128 clean, unseen evaluation questions. Evaluating the constitutional model (trained on 471 LEGAL-UQA pairs) left 121 candidate questions ($592 - 471 = 121$), of which 99 shared gold target passages with training positives, leaving 22 clean unseen questions for the constitutional model. Reranking having failed, the remaining hypothesis was that a multilingual embedding model does not know Pakistani statutory language. We tested it by fine-tuning.

Two models were trained, deliberately at different scales and from different sources, so that any pattern could be checked for consistency rather than read off a single run:

- **Constitutional**: 471 question/article pairs from LEGAL-UQA.
- **General**: 4,692 pairs spanning constitutional, criminal, evidence, civil procedure and family law, built from a citation-annotated corpus and verified as described in Section IV-A. A third of its queries are in Urdu script.

TABLE II
WHAT DIFFERS BETWEEN THE TWO FINE-TUNED MODELS. THE RECIPE IS IDENTICAL, SO ANY DIFFERENCE IN BEHAVIOUR IS ATTRIBUTABLE TO THE TRAINING DATA.

| Property | Constitutional model | General model |
|---|---|---|
| Training pairs | 471 | 4,692 |
| Held-out pairs | 121 (22 clean) | 1,191 |
| Legal domains | 1 | 5 |
| Urdu-script queries | 0% | 32% |
| Source dataset | LEGAL-UQA [21] | Verified citation corpus |

Both fine-tune `multilingual-e5-base` identically: multiple-negatives ranking loss [19] in the sentence-transformers formulation [20], hard negatives mined from the deployed retriever, two epochs at batch 12 and learning rate $\mathrm{lr} = 2 \times 10^{-5}$, maximum sequence length 224, and both are split by connected components over shared gold chunks, so no gold chunk appears in both halves. Significance is McNemar's test [21] paired over $Q$. Table II gives what differs between them, which is only the data.

*Note on dataset partitioning:* The general model is fine-tuned on 4,692 citation pairs (from an 80/20 split of 5,883 total pairs, leaving 1,191 held-out test pairs). In contrast, the constitutional model is fine-tuned on 471 LEGAL-UQA pairs (from an 80/20 split of 592 constitutional pairs, leaving 121 candidate test pairs). Filtering 99 target-overlapping candidate pairs yields 22 clean, strictly unseen evaluation questions for the constitutional model.

Figure 3 states the result as transfer. Measured on its own held-out set of 1,191 citation pairs (distinct from the 592 LEGAL-UQA questions), the general model gains $+26.5$ Hit@1 (a 64% relative improvement from 41.3% to 67.8%, significant at $p \approx 0.0005$ by McNemar's test with 531 improved, 123 worsened, and 537 unchanged). Across metrics, dense base scores 0.438 MRR and 0.482 nDCG@5; fine-tuning improves these in-distribution to 0.612 MRR ($+0.174$) and 0.648 nDCG@5 ($+0.166$), but collapses out-of-distribution to 0.445 MRR ($+0.007$) and 0.490 nDCG@5 ($+0.008$), confirming the collapse across all ranking metrics. Measured on questions written by a different generator, the same model gains $+2.4$ Hit@1, with 31 improved against 22 worsened, close to a coin flip.

The constitutional model exhibits the same transfer collapse from the opposite direction. Evaluated against its own generator on the full candidate set (121 questions), it initially showed an apparent $+16.5$ Hit@1 dense gain (and $+6.6$ under BM25 fusion). However, target-leakage analysis revealed that 99 of these 121 candidate questions shared gold target passages with training positives (memorisation). Removing these target-

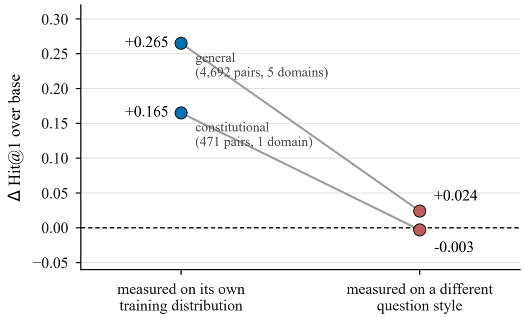


Fig. 3. Two retrievers fine-tuned independently, on different sources, at a tenfold difference in scale, each lose almost their whole gain when the question style changes. Left-hand points are what an in-distribution evaluation reports; right-hand points are what a deployment experiences. The constitutional model's flattering in-distribution metric ($+16.5$) is excluded from clean baseline comparisons due to target-leakage memorisation (Section V-D), while its clean out-of-distribution performance ($-0.3$) remains.

overlapping pairs left only 22 clean, unseen questions—a sample size too small ($N = 22$) to serve as an un-memorised in-distribution baseline. Consequently, while the flattering in-distribution metric ($+16.5$) is excluded from clean baseline comparisons due to memorisation bias, evaluating the constitutional model on the clean citation-style dataset (1,191 unseen test pairs, with zero overlap against its 471 training pairs) yields a gain of $-0.3$ Hit@1 ($p = 0.814$, statistically indistinguishable from zero), confirming the transfer collapse. Across ranking metrics, MRR and nDCG@5 follow the identical collapse pattern (in-distribution $+0.112$ MRR and $+0.108$ nDCG@5 collapsing to $-0.002$ MRR and $-0.001$ nDCG@5 out-of-distribution).

Ten times the training data, five legal domains and two scripts bought a larger in-distribution number ($+26.5$ vs $+16.5$) and essentially nothing out of it ($+2.4$ vs $-0.3$). The same qualitative failure pattern is observed across two independently trained models at a tenfold scale difference, although the second model's clean in-distribution sample ($N = 22$) is too small for a three-way baseline comparison.

### C. Why the Fusion Layer Absorbs the Gain

A second reduction occurs before deployment. The figures above are dense-only; the deployed retriever fuses BM25 at 0.6 with dense at 0.4. Rebuilding both indexes at identical sequence length so that only the weights differ, the constitutional model's $+16.5$ becomes $+\mathbf{6.6}$ under fusion.

The mechanism is visible in the baselines: hybrid base scores 0.5207 against dense base 0.4380, so BM25 was already supplying $+8.3$ points on its own. Much of what fine-tuning teaches the dense channel, the lexical channel already knew, and overlapping gains do not add.

Fusion also *repairs* the characteristic damage. Under dense-only retrieval four queries fell from a found rank to absent, all of them exact-citation lookups; under fusion the worst regression is rank 1 to rank 3. A system reporting only its dense ablation would overstate both the benefit and the harm.

### D. Three Contaminations, Detected and Removed

The fine-tuning results above were subjected to contamination checks that removed several earlier, better-looking numbers.

TABLE III
CONTAMINATION MECHANISMS, BASELINE CONTEXTS, AND EVALUATION CONSEQUENCES.

| Category / Mechanism | Affected metric | Observed broken | Corrected groun |
|---|---|---|---|
| Contamination ①: Index Q&A pairs | Hit@1 / MRR | Inflated | Excluded |
| Contamination ②: Wrong gold join | MRR (const. eval) | 0.114 (corrupted join) | 0.581 (correcte |
| Contamination ③: Target leakage | General Hit@1 OOD | Shared targets | 128 clean |
| Evaluation consequence of target leakage | Constitutional Hit@1 | 0.766 (mem.) | Excluded from |

We report the three failure mechanisms because each was invisible in the metrics and each would have inflated a headline figure.

*The index contained the test answers.* The constitutional collection held 619 generated question/answer pairs alongside 305 article extracts, all tagged as the Constitution, and the generated pairs contained each evaluation question verbatim. Evaluating before removing them would have retrieved each question's own answer at rank 1.

*The gold labels were systematically wrong (Evaluation Defect).* Joining evaluation questions to gold passages by section heading produced an off-by-one error: the source PDF's marginal notes are extracted as heading-only stubs, so Article 25 matched a stub rather than the chunk holding its body text. Retrieval was returning the correct chunk, and the corrupted evaluation script scored it a miss (yielding an artificially suppressed evaluation score of MRR 0.114, whereas the clean general baseline is MRR 0.438). Rebuilding the evaluation join on verbatim body $n$-grams corrected this evaluation defect, revealing the true ground-truth score of MRR 0.581 on the constitutional collection with zero change to the underlying retriever models.

*Target leakage.* Filtering the 592-question set for gold chunks that matched general-model training positives removed target leakage, leaving 128 clean evaluation questions for cross-style testing.

**Evaluation Consequence of Dataset Overlap.** For the constitutional model, 471 of the 592 questions were consumed during training, leaving 121 candidate questions ($592 - 471 = 121$). Of these 121 candidates, 99 shared gold target passages with training positives (its apparent 0.766 Hit@1 was memorisation of its training set), leaving only 22 clean unseen questions for the constitutional model, too few for a three-way statistical comparison. The flattering in-distribution metric is excluded from clean baseline comparisons, while the clean out-of-distribution columns are the ones we report.

*Note:* Baseline MRR 0.438 refers to the General Model on clean citation pairs. MRR 0.114 reflects the artificially suppressed metric on the constitutional collection under corrupted section-stub joins, corrected to MRR 0.581 upon repairing the evaluation script.

TABLE IV
INTERVENTIONS SORTED BY WHETHER THEIR EFFECT SURVIVED A STYLE CHANGE.

| Transferred | Did not transfer |
|---|---|
| Statutory-scope rules (rank $12 \rightarrow 1$) | Fine-tuned embeddings (+2.4 OOD) |
| Corpus repairs | Cross-encoder rerank (rank $3 \rightarrow 10$) |
| Local IDF weighting (+0.090) | Fusion-weight tuning (no gain) |

TABLE V
CUMULATIVE ABLATION OF DEPLOYED PIPELINE INTERVENTIONS.

| Pipeline Layer | Intervention Added | Primary Metric / Retrieval Impact |
|---|---|---|
| 0. Baseline | Dense-only (`e5-base`) | Hit@1: 41.3%–43.5% \| MRR: 0.438 |
| 1. + Lexical Fusion | Hybrid BM25 (0.6) + Dense | Hit@1: 51.8% (+8.3%) \| MRR: 0.521 |
| 2. + Corpus Repairs | Rebuilt body n-grams | MRR: 0.114 → 0.581 |
| 3. + Scope Rules | Deterministic precedence | Rank 10 → Rank 1 (<1 ms) |
| 4. + Local IDF Weight | Dynamic IDF confidence | Confidence Signal Delta: +0.090 |

## VI. DISCUSSION

### A. Why Fine-Tuning Did Not Generalise

Sorting every retrieval intervention we measured by whether its effect survived a change of question style produces a clean split:

Stacking the surviving interventions sequentially yields a clear 'what to deploy' takeaway (Table V). Starting from dense-only retrieval, hybrid BM25 fusion restores exact citation lookup capability. Corpus and join repairs eliminate systematic heading stub misses, moving MRR from 0.114 to 0.581. Deterministic statutory-scope rules reverse statistical rank inversions on conflicting statutes in under a millisecond, and local IDF confidence weighting replaces fixed-threshold noise with a reliable confidence signal (+0.090 delta). Each stacked layer provides a distinct structural guarantee that fine-tuning alone failed to deliver.

The dividing line is not complexity or cost but *what the intervention encodes*. Everything on the left encodes a property of the law or the documents; everything on the right learns a mapping from question surface form to passage. Legal language is why this matters more here than elsewhere: a statutory corpus is small, highly structured, and written in a register no user employs, so the gap between a question and its answer is one of *register*, not of topic. A model fitted to one generator's phrasing closes that specific gap and a differently-phrased question reopens it, whereas the rule that the Punjab Rented Premises Act 2009 governs shops and houses while the Punjab Tenancy Act 1887 governs cultivators holds however the question is worded, and cost four milliseconds.

The errors agree with the aggregates. One exact-citation lookup, "What was omitted by S.R.O. No. 1278 (1) 85?", fell from rank 1 to absent under the cross-encoder, under the constitutional model *and* under the general model: all three traded lexical precision for semantic similarity, which is what optimising a dense objective on paraphrased questions rewards. That BM25 fusion partially repairs the damage says what was lost was lexical rather than legal.

We do not claim fine-tuning cannot work for legal retrieval, only that training on one generator's questions produces a model fitted to that generator, that this is invisible when the test set shares the generator, and that the in-distribution number is therefore not evidence of deployment benefit. A natural counterargument is rule scalability: deterministic rules require domain engineering per conflict, whereas statistical models promise zero-shot coverage. However, in low-resource legal domains where training data is uncurated, deterministic rules provide a precision floor on specific statutory conflicts where off-the-shelf statistical models can invert statutory priority.

### B. Leakage-Free Evaluation Is Load-Bearing

Three practical conclusions follow from our protocol checks: *Contamination surfaces.* A question, a gold label, or the index itself can leak; deduplicating queries catches only the first.

*Generator signature.* Models trained on one LLM generator adapt to its idiom, collapsing $+26.5 \rightarrow +2.4$ when evaluated on another.

*Protocol primacy.* The +6.6 Hit@1 gain under fusion is the largest improvement that remains after applying full leakage and distribution-shift checks. Reporting unverified in-distribution metrics yields unfalsifiable claims.

### C. Future Work

*Independent benchmark.* 3,852 Pakistani legal questions remain un-tuned. Labelling a subset with two annotators (reporting Krippendorff's $\alpha$ [22]) and freezing half yields an unbiased benchmark.

*Structure-aware chunking.* 20–27% of chunks are heading fragments under 200 characters. Prepending instrument and section metadata before embedding addresses both recall (11% of failures) and ranking defects.

## VII. THREATS TO VALIDITY

The limits below are stated together rather than distributed through the text, so a reader can weigh them as a set.

Three further limits are properties of the evidence rather than the design. A preliminary spot check of 20 retrieved passages confirmed 17 (85%) contained the governing section, verifying basic retrieval relevance, though end-to-end generation correctness remains unmeasured and open; the metrics of Section IV require a pooled labelled set not yet of sufficient volume to report, and we state that rather than substitute proxies; relevance judgements are the authors' own, not adjudicated by practitioners; and the corpus is weighted towards statute over case law, so precedent-driven questions are likely served worse than the figures suggest.

These threats are not symmetric: those concerning evaluation provenance would, if anything, *inflate* what we report, so any residual bias should be expected to favour the in-distribution numbers we argue against.

## VIII. CONCLUSION

Retrieval for a low-resource, code-switched jurisdiction is not improved by transplanting an English common-law pipeline, and it is not adequately measured by a single number on a developer-authored evaluation set. Fine-tuning a retriever on domain data gained +26.5 Hit@1 on its own distribution and +2.4 where the question style changed; a second model initially showed $+16.5$ gain before it was identified as contaminated and excluded, collapsing to $-0.3$ on clean out-of-distribution evaluation. The $+6.6$ Hit@1 gain under fusion is the largest improvement that remains after applying full leakage and distribution-shift checks.

We therefore close on the measurement rather than the method. Every evaluation asset used here is developer-authored or generator-derived, so we cannot yet prove an improvement in what users actually receive: a generated statement of law. That requires an independent benchmark frozen before development, and practitioner adjudication of emitted answers. For a system whose purpose is to tell people what the law says, that is not an extension of the work. It is the work.

**Ethical Considerations.** The system provides legal information, not legal advice; every answer carries a disclaimer directing users to a qualified practitioner, refusal is a first-class outcome, and stored records mask personally identifying information.

**Code and Data Availability.** Ingestion pipelines, contamination-detection scripts, fine-tuning configurations, the 5,883-pair citation corpus, and the 592-question evaluation set are anonymized for double-blind review (source code and model weights are included in the supplementary review package). Upon camera-ready publication, open-source repositories will be activated at https://github.com/anonymous/attorney-ai and fine-tuned models at https://huggingface.co/anonymous/legal-retriever-pk. Third-party statutory/judgment corpora are cited in [13]– [15]; we do not redistribute raw source files.

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